

methodological hierarchy for degenerate quantum eigenspaces

	residual PINN	orthogonality penalty	orthogonal projection	subspace Ritz	symmetry sectors
1D excited states	×	±		—	—
square degeneracy	×	±	±		—
rectangle splitting	?	?	±		—
disk $m = 1$ doublet	?	±	±		—
ellipse splitting	?	×	×	±	

success
partial/unstable
failure
not run/missing
not applicable